

CLC also includes the Walrus Service for S3-compatible storage.

Set up cluster controllers (CC):

Install CCs to handle networking and communication between the CLC and node controllers in multi-cluster environments, ensuring VMs are properly configured.

Node controller (NC)

configuration: Configure NCs to host virtual machines, manage local resources (CPU, memory, disk), and interact with the CLC.

Storage controller (SC) setup:

Install SCs to manage EBS-like volumes, handling storage requests and attaching volumes to VMs.

Configure networking: Set up networking using Eucalyptus modes (managed, unmanaged, or static). ‘Managed Mode’ handles IP addressing and networking automatically.

Testing and access: Test the cloud by launching instances via the CLI or web interface, ensuring proper VM provisioning and storage attachment.

Provisioning compute resources

Eucalyptus allows on-demand provisioning of virtual machines (VMs), similar to AWS EC2 instances.

Cloud controller (CLC): CLC is central to Eucalyptus, handling user requests and managing resource allocation. It supports interactions via APIs (compatible with AWS EC2), web interfaces, or command-line tools, enabling users to request VMs, configure networks, and allocate storage.

Node controller (NC): The node controller (NC) runs on physical servers, managing VMs by allocating compute, memory, and storage resources using hypervisors like KVM or Xen. It interacts with the cloud controller (CLC) and cluster controller (CC) to fulfil resource requests.

Storage controller (SC): The storage controller (SC) provides persistent block storage (EBS-like) to VMs. It manages volumes that retain

data beyond VM termination and coordinates with the CLC for efficient storage request handling.

Here are the steps that must be followed to provision compute resources.

Define virtual machine image:

Choose or upload a VM image (e.g., EMI or AMI) from the catalogue.

Launch VM instance: Use the ‘Eucalyptus Management Console’ or command-line tools to specify instance type and network settings, which the NC uses to provision the VM.

Monitor and manage VMs: Access running VMs via SSH or remote desktop, and use Eucalyptus tools to monitor performance metrics like CPU usage and memory consumption.

Managing storage resources

Eucalyptus supports block storage and object storage.

Block storage (EBS-like volumes):

Managed by the storage controller, block storage provides persistent volumes similar to AWS EBS. Users can create, attach, and detach volumes, with features like snapshots for backups.

Object storage (S3-compatible):

The Walrus Service offers S3-compatible storage, enabling users to manage data objects with features like buckets, ACLs, and versioning. It provides scalable storage for unstructured data, with integration into IAM for access control.

Overall, Eucalyptus offers a robust and cost-effective AWS-compatible solution for managing compute and storage resources in private and hybrid clouds.

Integrating Eucalyptus with AWS

Eucalyptus facilitates hybrid cloud integration by offering compatibility with AWS APIs. Organisations can run a private cloud on their hardware with Eucalyptus while leveraging AWS for additional capacity during peak times or for specific services.

EC2 (Elastic Compute Cloud):

Eucalyptus provides an EC2-compatible

environment for managing virtual machines, easing the transition between private and public clouds.

S3 (Simple Storage Service): The Walrus service in Eucalyptus offers S3-compatible storage, allowing for seamless data synchronisation and backup between on-premises and AWS.

IAM (identity and access management): Eucalyptus supports IAM-compatible policies, aligning access controls across both private and public cloud environments.

Seamless integration: Eucalyptus ensures that applications and data can easily move between private and public clouds. For example, apps can be developed and tested in Eucalyptus and then deployed to AWS for production. Businesses can also use AWS for additional resources during peak times, while maintaining baseline operations in their private cloud.

Use cases and real-world applications

Disaster recovery: Eucalyptus can be used to maintain backups in AWS, ensuring quick recovery and continuity in case of a disaster.

Development and testing:

Developers can use Eucalyptus for testing in an EC2-compatible environment, with the option to deploy to AWS for production.

Extending on-premises resources:

Organisations can scale their on-premises resources with AWS during high-demand periods, using Eucalyptus for seamless integration.

NASA Nebula used Eucalyptus for a hybrid cloud setup, managing private cloud infrastructure while leveraging AWS. UC Santa Barbara implemented Eucalyptus for research computing, extending to AWS for larger-scale tests as needed.

Benefits of using Eucalyptus

Cost efficiency: Eucalyptus reduces costs by enabling organisations to use their existing infrastructure for private